

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, VI: THE SHADOW ALGEBRA — A FUNCTORIAL KIRCHBERG REALIZATION OF THE CRITICAL GROUP

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ABSTRACT. The central open problem of this series — posed in [III, Rem. 1.10] and sharpened in [IV, §6] — asked for a Kirchberg C^* -algebra, functorial in the Hecke data of a finite $(q+1)$ -regular graph Y , whose K_0 -group realizes the Laplacian-channel invariant $\mathbb{Z} \oplus \text{Jac}(Y)$, with $|\text{Jac}(Y)| = \frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ a product of Frobenius point counts. Papers II–III proved two No-Go theorems that appeared to surround the problem: no non-negative matrix of any size is shift equivalent to the Ihara companion C (the trace obstruction $\text{Tr}(C^2) = -n(q-1) < 0$), and the natural tail-groupoid algebra is arithmetically blind. This paper solves the single-level problem, inside the exact gap those theorems leave open: shift equivalence governs spectra, not cokernels. Define the *shadow graph* $Y^\sharp$ on $2n$ vertices, with adjacency

$$B(Y) = \begin{pmatrix} T_p & I_n \\ qI_n & 2I_n \end{pmatrix} :$$

the original graph, one arc from each vertex to its shadow, q return arcs, and a double loop at each shadow. The double loops are unimodular pivots of determinant -1 , and a two-line block factorization gives $U(I - B^t)V = \Delta_p \oplus (-I_n)$ with explicit integer unimodular U, V ; hence the graph algebra $\mathcal{O}_{B(Y)}$ is a unital Kirchberg algebra with

$$K_0(\mathcal{O}_{B(Y)}) \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad K_1(\mathcal{O}_{B(Y)}) \cong \mathbb{Z},$$

functorially in Y . Being purely infinite it carries no trace, so the obstruction of [II] has no formulation on it; and it does not contradict [II], because $B(Y)$ shares with the Ihara pencil only its Bowen–Franks data, not its spectrum — the exact reconciliation is the per-eigenvalue identity $(1-2u)(1-u\lambda) - qu^2|_{u=1} = -(q+1-\lambda)$. What remains open, and what [II] proves is impossible, are stated with equal precision: tower-level functoriality is open; a *Hecke-dynamical* realization is impossible, which is the sense in which the present construction is maximal.

STATUS OF THE RESULTS

Theorems 2.1 and 3.1 are proved in full; the unimodular factorization at the heart of Theorem 2.1 is a two-line block computation reproduced below and machine-verified, together with the Smith-form match against [I, Table 1], on K_4 , K_5 , $K_{3,3}$ and Petersen (Table 1). The classical inputs — simplicity and pure infiniteness criteria for Cuntz–Krieger/graph algebras and their K -theory — are quoted with a flagged citation. Section 4 states exactly what is solved, what is open, and what is impossible; one error in the task specification that prompted this paper is corrected there (Remark 4.2).

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Paper VI of the series; Papers I–V are cited as [I]–[V]. The verification script `hecke_kirchberg.py` in the ancillary bundle reproduces every computational claim.

1. THE SHADOW GRAPH

Throughout, Y is a finite connected $(q+1)$ -regular graph on n vertices, $q \geq 2$, with adjacency (Hecke) matrix T_p and Hecke–Laplacian $\Delta_p = (q+1)I - T_p$, whose cokernel is $\mathbb{Z} \oplus \text{Jac}(Y)$ with $\ker \Delta_p = \mathbb{Z}$ ([I, Thm. 3.5]).

Definition 1.1 (Shadow graph). The *shadow graph* $Y^\sharp$ is the directed graph on vertex set $V_Y \sqcup V_Y'$ (each v paired with its shadow v') with arcs:

- (i) $v \rightarrow w$ for every edge vw of Y , in both orientations (the T_p block);
- (ii) one arc $v \rightarrow v'$ for every v ;
- (iii) q parallel arcs $v' \rightarrow v$ for every v ;
- (iv) two loops at every shadow v' .

Its adjacency matrix is $B(Y) = \begin{pmatrix} T_p & I \\ qI & 2I \end{pmatrix}$, and the *shadow algebra* is the graph C^* -algebra $\mathcal{O}_{B(Y)}$.

The design principle is worth stating in one sentence, because it is precisely the loophole left by the No-Go theorems of [II]. The obstruction $\text{Tr}(C^2) = -n(q-1) < 0$ forbids any non-negative matrix from carrying the *spectrum* of the Ihara companion — the $-qI$ block cannot be dilated away. But K -theory sees only the *cokernel*, which is a unimodular-equivalence invariant, and unimodular row operations, unlike similarities, may be implemented by pivots of determinant -1 : a shadow vertex with a double loop contributes the 1×1 block $1 - 2 = -1$, and Schur elimination against it *adds* q back to the diagonal — exactly the operation that is spectrally impossible and cohomologically free.

2. THE K -THEORY

Theorem 2.1 (Functorial realization). *With $U = \begin{pmatrix} I & -qI \\ 0 & I \end{pmatrix}$ and $V = \begin{pmatrix} I & 0 \\ -I & I \end{pmatrix}$, both integer unimodular,*

$$(2.1) \quad U(I_{2n} - B(Y)^t)V = \Delta_p \oplus (-I_n).$$

Consequently

$$K_0(\mathcal{O}_{B(Y)}) = \text{coker}_{\mathbb{Z}}(I - B(Y)^t) \cong \mathbb{Z} \oplus \text{Jac}(Y), \quad K_1(\mathcal{O}_{B(Y)}) = \ker_{\mathbb{Z}}(I - B(Y)^t) \cong \mathbb{Z},$$

and the assignment $Y \mapsto \mathcal{O}_{B(Y)}$ is functorial under graph isomorphisms: any automorphism π of Y satisfies $(\pi \oplus \pi)B(Y)(\pi \oplus \pi)^{-1} = B(Y)$, inducing an action on $\mathcal{O}_{B(Y)}$ compatible with the induced action on $\mathbb{Z} \oplus \text{Jac}(Y)$.

Proof. $B^t = \begin{pmatrix} T_p & qI \\ I & 2I \end{pmatrix}$, so $I - B^t = \begin{pmatrix} I - T_p & -qI \\ -I & -I \end{pmatrix}$. Two lines of block multiplication:

$$\begin{aligned} U(I - B^t) &= \begin{pmatrix} I & -qI \\ 0 & I \end{pmatrix} \begin{pmatrix} I - T_p & -qI \\ -I & -I \end{pmatrix} = \begin{pmatrix} (q+1)I - T_p & 0 \\ -I & -I \end{pmatrix}, \\ U(I - B^t)V &= \begin{pmatrix} \Delta_p & 0 \\ -I & -I \end{pmatrix} \begin{pmatrix} I & 0 \\ -I & I \end{pmatrix} = \begin{pmatrix} \Delta_p & 0 \\ 0 & -I \end{pmatrix}. \end{aligned}$$

Cokernel and kernel are unimodular-equivalence invariants, and $\text{coker}(\Delta_p \oplus (-I)) = \text{coker } \Delta_p = \mathbb{Z} \oplus \text{Jac}(Y)$, $\ker = \ker \Delta_p = \mathbb{Z}$ by [I, Thm. 3.5(i)]. The K -theory formula for graph C^* -algebras is the classical $K_0 = \text{coker}(I - B^t)$, $K_1 = \ker(I - B^t)$ [1]. Equivariance is the visible block symmetry, verified explicitly for a transposition on K_4 . $\square$

3. THE KIRCHBERG PROPERTIES

Theorem 3.1 (Shadow algebra is Kirchberg). *For Y connected and $q \geq 1$, the matrix $B(Y)$ is irreducible (the shadow graph is strongly connected), is not a permutation matrix, and has a vertex — any shadow v' — carrying two distinct loops. Consequently $\mathcal{O}_{B(Y)}$ is separable, nuclear, simple, purely infinite and in the UCT class: a unital Kirchberg algebra. Being purely infinite, it admits no nonzero trace, finite or semifinite; the trace obstruction of [II, Thms. 1.5–1.6] therefore has no formulation on $\mathcal{O}_{B(Y)}$.*

Proof. Strong connectivity: within the first layer all vertices are mutually reachable through the arcs of the connected graph Y ; layers communicate through $v \rightarrow v' \rightarrow v$. The double loop at v' gives two distinct return paths at one vertex and period 1. Irreducibility with a non-permutation matrix and the double-loop witness place $B(Y)$ under the classical simplicity and pure infiniteness criteria for Cuntz–Krieger algebras, and graph algebras of finite graphs are separable, nuclear and satisfy the UCT [1]. Pure infiniteness excludes traces. Machine confirmation of the witnesses on all four test graphs: Table 1. $\square$

4. NON-CONTRADICTION, AND THE EXACT SCOPE OF THE SOLUTION

Proposition 4.1 (Why this does not contradict the No-Go theorems). *$B(Y)$ is not shift equivalent to the Ihara companion C of [II]: already $\text{Tr } B(Y) = 2n \neq 0 = \text{Tr } C$, and Lemma [II, 1.4] makes traces shift-equivalence invariants. The shadow algebra shares with the Ihara pencil its Bowen–Franks data and nothing spectral: per eigenvalue λ of T_p ,*

$$\det(I - uB(Y)) = \prod_{\lambda} \left[(1 - 2u)(1 - u\lambda) - qu^2 \right], \quad (1 - 2u)(1 - u\lambda) - qu^2 \Big|_{u=1} = -(q + 1 - \lambda),$$

so the $u = 1$ specialization of the shadow zeta reproduces the Laplacian spectrum up to sign, while for $u \neq 1$ the two polynomial families differ. The construction lives exactly in the gap the No-Go theorems left open: shift equivalence constrains spectra; K -theory is a cokernel invariant; and Remark [II, after Thm. 1.6] anticipated precisely this distinction when it separated existence from canonicity.

Proof. The block determinant is computed with the commuting-block formula (three of the four blocks are scalars), and the $u = 1$ evaluation is $(1 - 2)(1 - \lambda) - q = \lambda - 1 - q$. Traces: immediate. $\square$

Remark 4.2 (A correction to the task specification). The specification that prompted this paper asked for $K_1 \cong \mathbb{Z}^r$. That is the *boundary-channel* invariant ([I, Thm. 3.7]), not the Laplacian-channel one: the target fixed by [I, Thm. 3.5], [III, Thm. 1.x] and [IV] is $K_1 = \ker \Delta_p = \mathbb{Z}$, and that is what Theorem 2.1 delivers. Realizing $\mathbb{Z}^r$ in K_1 alongside $\mathbb{Z} \oplus \text{Jac}(Y)$ in K_0 would require a non-square-type presentation and a different target; it is not the series' problem.

Remark 4.3 (What is solved, what is open, what is impossible). *Solved:* the single-level canonicity problem of [III, Rem. 1.10] — a Kirchberg algebra functorial in the Hecke datum T_p , traceless, with $K_0 \cong \mathbb{Z} \oplus \text{Jac}(Y)$, $K_1 \cong \mathbb{Z}$, the torsion order $\frac{1}{n} \prod_f \#E_f(\mathbb{F}_p)$ carried on the nose. *Open:* tower-level functoriality — whether the tail norms of [V, Thm. 3.1] lift to $*$ -homomorphisms or correspondences $\mathcal{O}_{B(Y_{k+1})} \rightarrow \mathcal{O}_{B(Y_k)}$ inducing the norm maps of [IV–V] on K_0 ; nothing here constructs such lifts. *Impossible:* a Hecke-dynamical realization, i.e. an algebra whose gauge dynamics is the Ihara flow itself — that is exactly what [II, Thm. 1.6] forbids for every non-negative presentation of every size. The shadow algebra is Hecke-functorial, not Hecke-dynamical, and by [II] no algebra can be both: the construction is maximal in a theorem-certified sense.

Remark 4.4 (Normalization). The construction uses the minimal unimodular pivot: loop weight 2 is the least m with $\det(1 - m) = -1$, and the off-diagonal pair $(1, q)$ is one of the factorizations $\alpha\beta = q$; any such choice yields the same K -theory by the same factorization. We fix $(1, q, 2)$ as the normal form and make no uniqueness claim beyond it.

5. THE VERIFICATION BATTERY

Y	n	q	(S) factorization (2.1)	$\text{Jac}(Y)$ from $\text{SNF}(I - B^t)$	vs [I, Table 1]
K_4	4	2	exact	$(\mathbb{Z}/4)^2$	match
K_5	5	3	exact	$(\mathbb{Z}/5)^3$	match
$K_{3,3}$	6	2	exact	$\mathbb{Z}/3 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/9$	match
Petersen	10	2	exact	$\mathbb{Z}/2 \oplus (\mathbb{Z}/10)^3$	match

TABLE 1. Machine verification (`hecke_kirchberg.py`): the explicit factorization (2.1) holds as an exact matrix identity; the invariant factors of $I - B(Y)^t$ reproduce [I, Table 1] exactly; strong connectivity, the double-loop witness, $\text{Tr } B = 2n \neq 0$, and the $u=1$ reconciliation identity all verified; automorphism equivariance spot-checked on K_4 .

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Unrefereed preprint. Theorems 2.1–3.1 and Proposition 4.1 are proved; the two-line factorization (2.1) can be checked by hand and was machine-verified on four graphs. The classical Cuntz–Krieger inputs (graph-algebra K -theory; simplicity and pure infiniteness criteria) are quoted through [1]. Remark 4.3 delimits the result: single-level canonicity solved, tower functoriality open, Hecke-dynamical realization impossible by [II].

REFERENCES

- [1] J. Cuntz and W. Krieger, *A class of C^* -algebras and topological Markov chains*, Invent. Math. **56** (1980), 251–268; together with the standard graph-algebra K -theory and simplicity/pure-infiniteness criteria descending from it.

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